

Design and Fabrication of Mach-Zehnder Interferometer and Michaelson Interferometer

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Abstract—Placeholder

I. INTRODUCTION

THE The Mach-Zehnder Interferometer (MZI) is a foundational photonic circuit that senses the relative phase shift between two beams split from a single source. A phase shift is obtained by creating a difference in the optical path length (OPL) between the two arms of a MZI through a physical path length difference, materials with different refractive indexes, temperature through the thermo-optic effect, etc.. A Michaelson interferometer (MI) is similar to a MZI but the beam is reflected by a waveguide mirror and travels through the original path difference achieving a OPL of $2\beta\Delta L$ in a compact area.

II. THEORY

The circuit begins with a y-splitter. A beam of light with input intensity I_i and electric field E_i . The outputs have an intensity of $I_1 = I_2 = I_i/2$ and the E-field $E_1 = E_2 = E_i/\sqrt{2}$. The combiner will have $E_1 = E_{i1}/\sqrt{2}$ and $E_2 = E_{i2}/\sqrt{2}$ which will result in $E_o = \frac{E_{i1}+E_{i2}}{\sqrt{2}}$. A plane wave is considered:

$$E_{i1} = \frac{E_1}{\sqrt{2}} e^{-\beta_1 L_1}$$

$$E_{i2} = \frac{E_2}{\sqrt{2}} e^{-\beta_1 L_2}$$

So the E-field output of the combiner is:

$$E_o = \frac{E_{i1} + E_{i2}}{\sqrt{2}} = \frac{E_i}{2} (e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2})$$

And the output intensity can be calculated as follows if the paths any differ from a path length:

$$I_o = \frac{I_i}{4} |e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}| = \frac{I_i}{4} (1 + \cos(\beta\Delta L))$$

Where $\Delta L = L_2 - L_1$.

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III. MODELING AND SIMULATION

A. Waveguide

This design uses as trip waveguide with a width of $w = 500nm$ and a height of $h = 220nm$. The TE and TM waveguide mode profiles are shown in the two following figures, respectively.

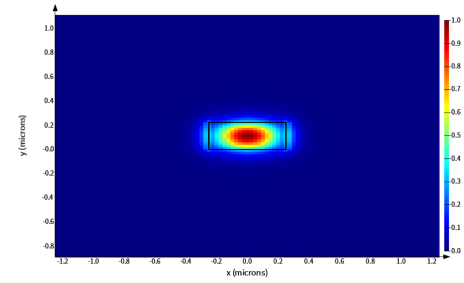


Fig. 1. Strip Waveguide TE Mode.

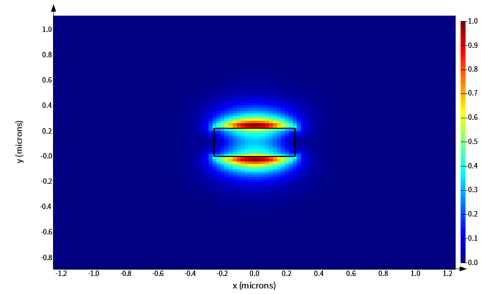


Fig. 2. Strip Waveguide TM Mode.

The E-field in Figure 2 penetrates deeper into the cladding making it more attractive for applications where high coupling is desired.

Using a second order model is calculated for the effective index in MATLAB.

$$n_{eff}(\lambda) = 2.4502 - 1.1321(\lambda - \lambda_o) - 0.0413(\lambda - \lambda_o)^2$$

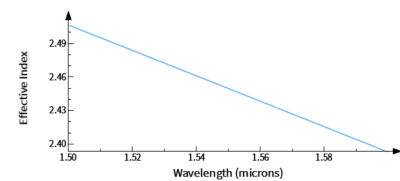


Fig. 3. Effective Index Against Wavelength

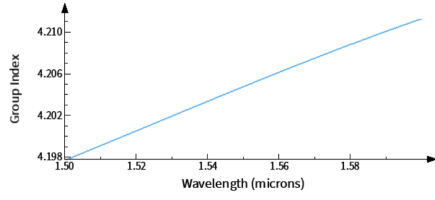


Fig. 4. Effective Index Against Wavelength

B. MZI/MI

The transfer function of the MZI is obtained:

$$\frac{I_o}{I_i} = \frac{1}{2}[1 + \cos(\beta\Delta L)]$$

The FSR of the MZI can be written as

$$FSR_{MZI} = \frac{\lambda^2}{n_g \Delta L}$$

And the MI can be written as

$$FSR_{MI} = \frac{\lambda^2}{n_g 2\Delta L}$$

The MZI and MI is simulated in Lumerical INTERCONNECT to obtain its spectra. The path length difference in the following figures is chosen to be $100\mu m$.

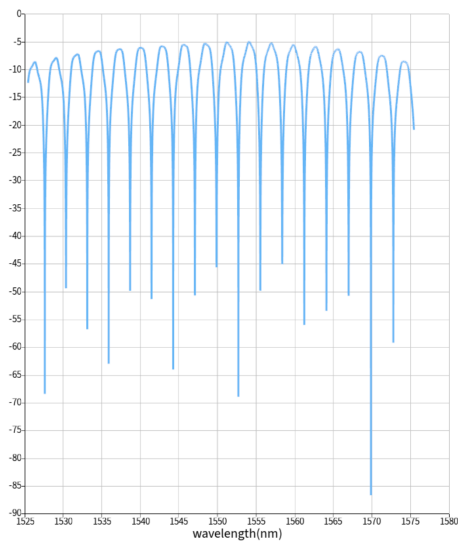


Fig. 5. 100um MI Spectra

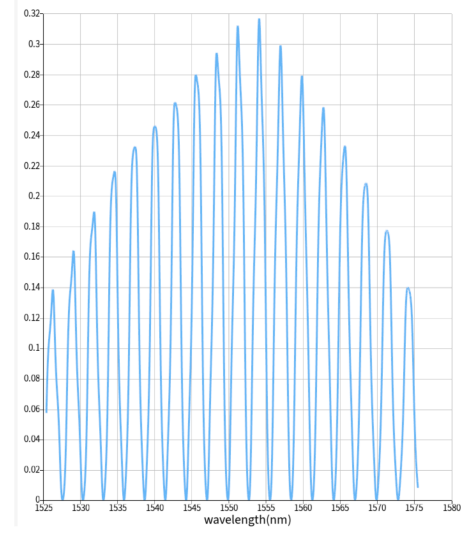


Fig. 6. 100um MI Spectra

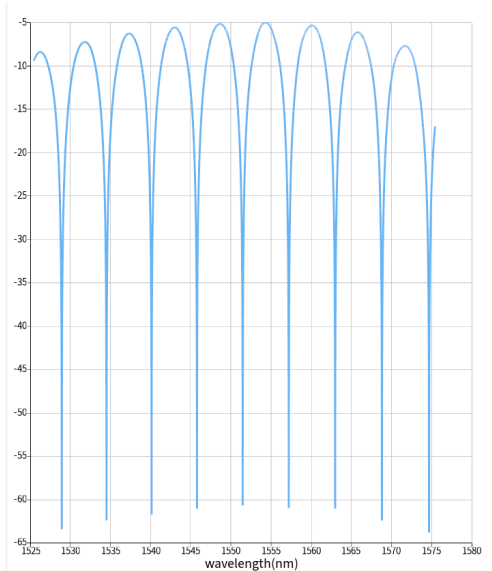


Fig. 7. 100um MI Spectra

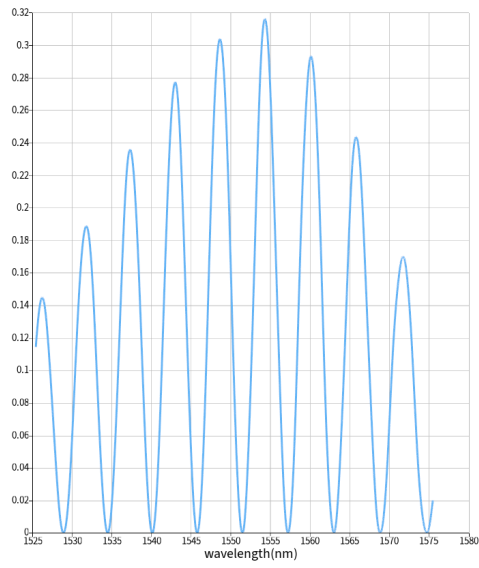


Fig. 8. 100um MI Spectra